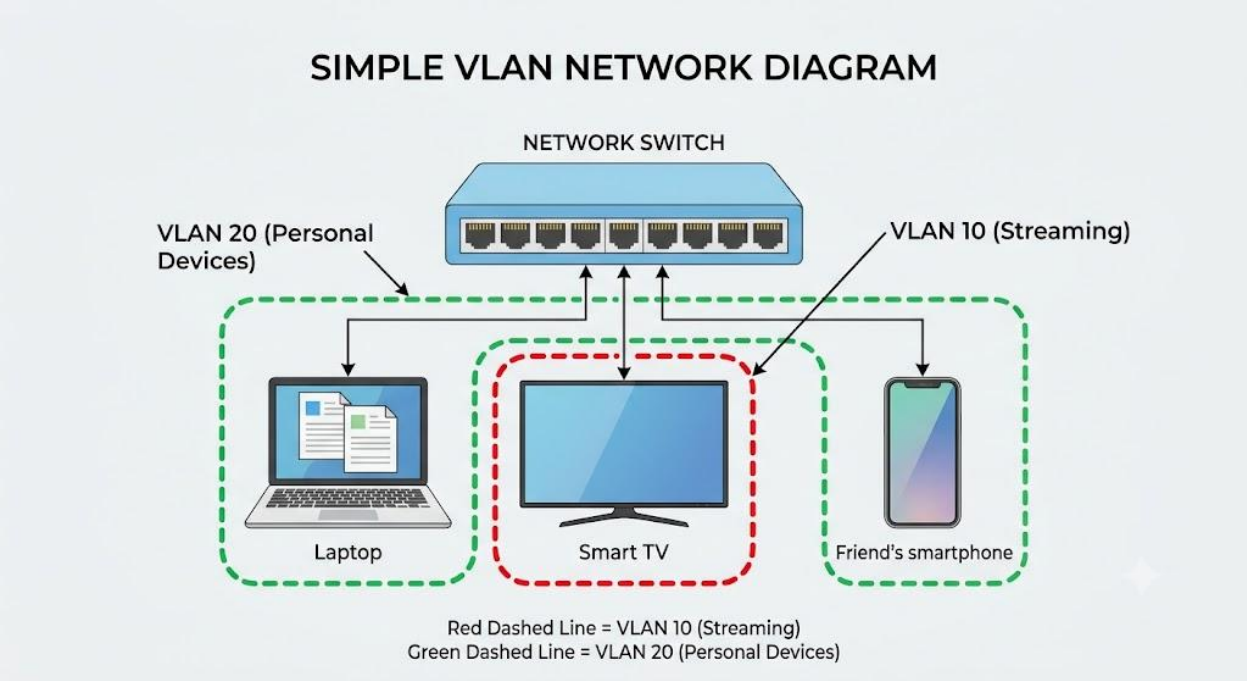
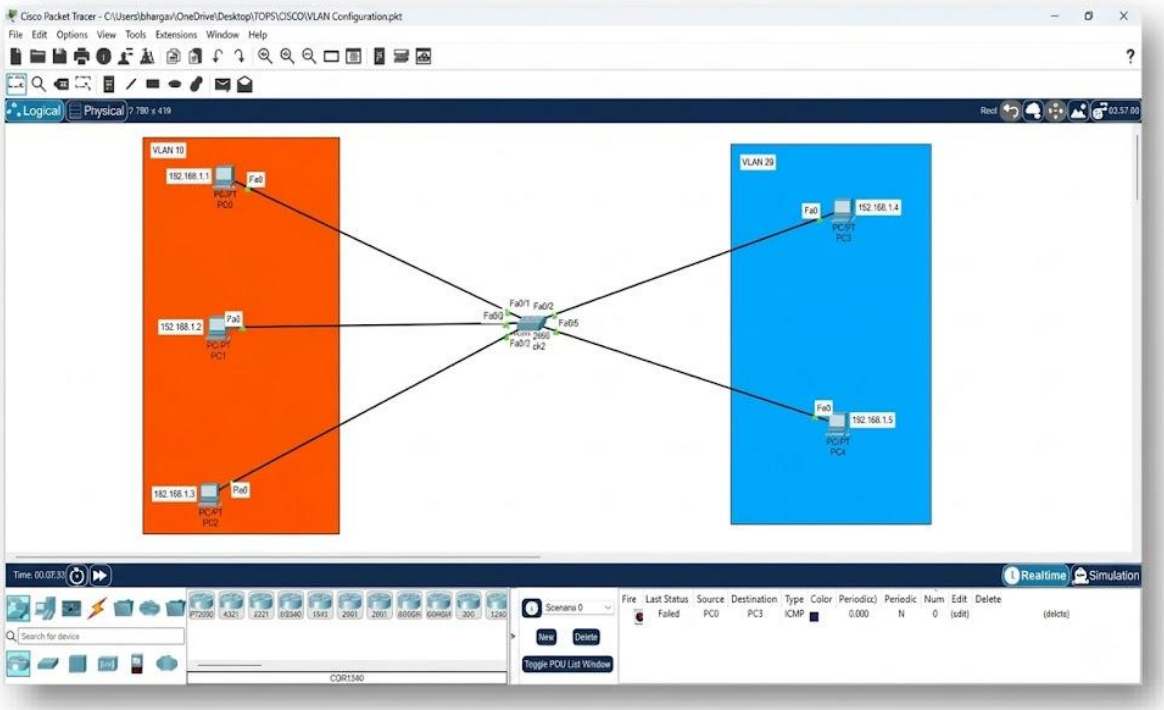
SWITCHING and VLANS

1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

Ans:-




Cisco Packet Tracer - C:\Users\bhargav\OneDrive\Desktop\TOPS\CISCO\VLAN Configuration.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
VLAN 10
VLAN 29
192.168.1.1
Fa0
PC0
192.168.1.2
Fa0
PC1
192.168.1.3
Fa0
PC2
192.168.1.4
Fa0
PC3
192.168.1.5
Fa0
PC4
Fa0/1
Fa0/2
Fa0/3
Fa0/5
Fa0/7
2960
sh2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Periodic(c)
Periodic
Num
Edit
Delete
Failed
PC0
PC3
ICMP
0.000
N
0
(edit)
(delete)

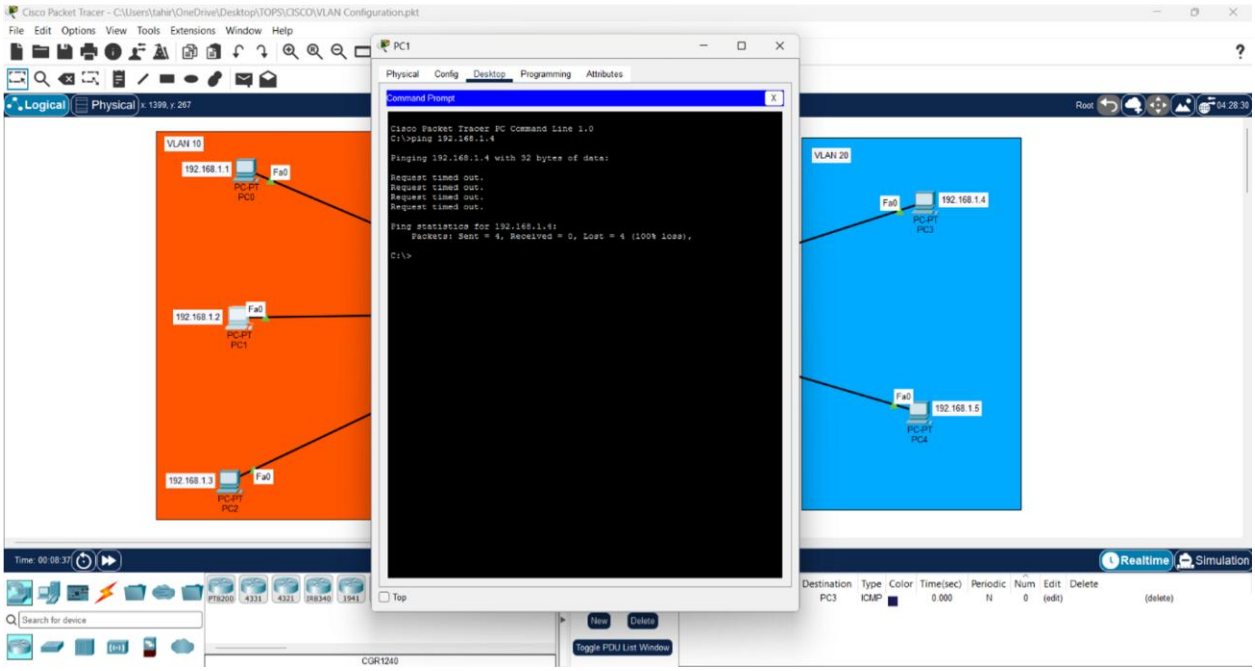
2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

Ans:-

Created VLAN 10 and VLAN 20 on the switch.

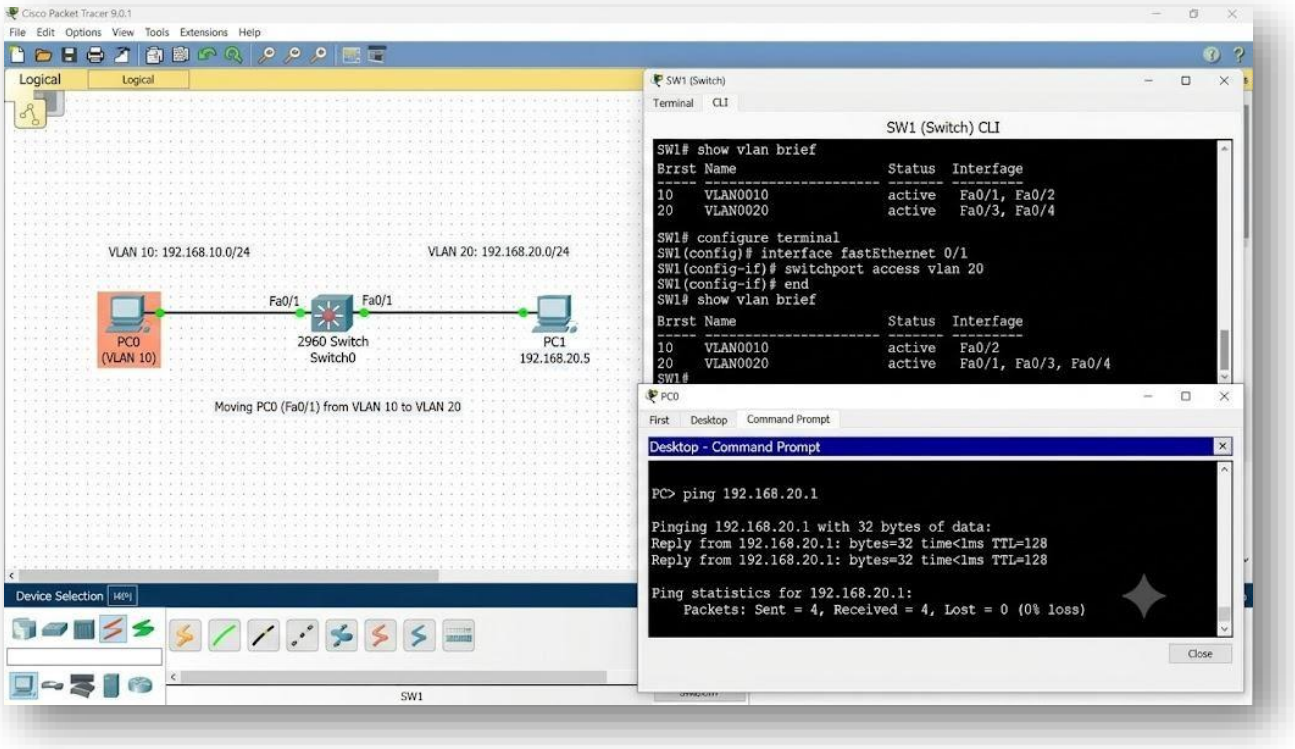
Used the ping command to test connectivity.



First, created two VLANs, VLAN 10 and VLAN 20, on the switch using the vlan 10 and vlan 20 commands. Then assigned three PCs to VLAN 10 and two PCs to VLAN 20 using the switchport access vlan command. After that, configured IP addresses for all PCs and checked the connection using the ping command. PCs in the same VLAN were able to ping each other, but PCs in different VLANs could not ping each other.

3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Ans:-



4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

Ans:-

Open Cisco Packet Tracer and select the switch.

Open CLI.

Enter these commands:

Switch> enable Switch# configure terminal Switch(config)# vlan 30 Switch(config-vlan)# exit Switch(config)# interface range FastEthernet 0/1 - 10 Switch(config-if-range)# switchport mode access Switch(config-if-range)# switchport access vlan 30 Switch(config-if-range)# exit Switch(config)# end Switch# show vlan brief

What happened?

I created VLAN 30 and used interface range to select FastEthernet 0/1 to 0/10 at once. Then I assigned all 10 ports to VLAN 30 without typing the commands separately for every port.